DESIGN BASIS REPORT

Meridian Data Centre Campus, Phase 1

Phase 1 Campus Build - 20MW IT Load

Client:	Meridian Digital Infrastructure Pvt Ltd.
EPC Contractor:	YOU Infrastructure
Consultant:	Apex Engineering Consultants
Location:	Navi Mumbai, India
Date:	January 15, 2026
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1. Executive Summary

This Design Basis Report (DBR) establishes the engineering criteria and parameters for the construction of Phase 1 of the Meridian Data Centre Campus, a state-of-the-art facility designed to support high-density IT operations in Navi Mumbai. The facility is designed to meet Uptime Institute Tier III concurrent maintainability requirements, allowing any component of the electrical or mechanical infrastructure to be taken out of service for maintenance without impacting the critical IT load.

2. Key Project Identity & Stakeholders

- **Project Title:** Meridian Data Centre Campus, Phase 1
- **Location:** Navi Mumbai, India, Maharashtra, India (coastal tropical environment)
- **Owner / Developer:** Meridian Digital Infrastructure Pvt Ltd.
- **EPC Contractor:** YOU
- **Lead Engineering Consultant:** Apex Engineering Consultants

3. Design Basis Parameters

Design Parameter	Target Criteria	Redundancy / Basis
Total IT Load	20000 kW (20 MW)	Max continuous IT operating power
UPS Topology	Static Online Double Conversion	N+1 per Group
Design Ambient Temperature	45°C Dry Bulb	Extreme summer condition, Navi Mumbai
Target Power Usage Effectiveness (PUE)	≤ 1.45	At full design load
Standby Generator Autonomy	48 Hours	Full campus load storage capacity
Cooling Unit Redundancy	Chilled Water Perimeter CRAH	N+1 per Hall
Battery Autonomy (UPS)	10 Minutes	At 100% rated module load

4. Architectural & Structural Limits

The facility is designed for extreme floor loadings to accommodate the heavy weights of double-conversion UPS cabinets and high-rate lead-acid battery cabinets. The structural concrete slab is engineered for a maximum static load limit of **2,800 kg/m²** in the UPS and battery rooms. Battery cabinets exceeding this limit are prohibited.

5. Environmental Design Parameters

Navi Mumbai is characterized by high summer ambient temperatures and high relative humidity (coastal monsoon climate). The cooling infrastructure is sized for a maximum outdoor design dry-bulb temperature of **45°C**.

Data halls shall be maintained at **23°C ± 2°C** supply air temperature (cold aisle containment) and relative humidity of 20-80% non-condensing to prevent electrostatic discharge or moisture condensation on server boards.